Preview: set-01/by_level/Grade_03-04_Ecolier/y2023_q19 (v1)

Write down every possible order in a system (who is first, then second, ...) and cross out the ones that break a rule

How to think about it: count the orders

When you have to count orders, do not guess. **Write down every order**, in a system, and then cross out the ones that break a rule.

- **A system:** first all orders where Hermann is first, then all where Felix is first, then all where Clemens is first. For 3 children there are 3 × 2 = **6 orders**.
- **Check every order with every rule.** One order can break two rules at the same time.

The contest question

Hermann is not the first. Felix is not the second. Clemens is not the third.

first	second	third	breaks a rule?
Hermann	Felix	Clemens	yes: Hermann is not the first; Felix is not the second; Clemens is not the third
Hermann	Clemens	Felix	yes: Hermann is not the first
Felix	Hermann	Clemens	yes: Clemens is not the third
Felix	Clemens	Hermann	**no - this order works**
Clemens	Hermann	Felix	**no - this order works**
Clemens	Felix	Hermann	yes: Felix is not the second

Two orders are left: Felix, Clemens, Hermann and Clemens, Hermann, Felix. The answer is **2**.

Watch out

- "3 rules, so 6 – 3 = 3" is wrong: a rule does not cross out just one order.
- "Each rule crosses out 2 orders, so 6 – 6 = 0" is wrong too: some orders break two rules, so you would cross them out twice.
- With 4 children there are 4 × 3 × 2 = 24 orders. Then start with the strongest rule ("... is the first", "... comes right after ...") so you have less to write.

Warm-up: small steps first (no points)

W1. Sara and Felix stand in a row. How many different orders are there?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

W2. Anna, Mia and Max stand in a row. Anna is not the third. How many different orders are there?

A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

W3. Hermann, Felix and Clemens enter a room one after the other. Hermann is not the first. Felix is not the second. Clemens is not the third. In which order can they enter?

A. Clemens, Felix, Hermann ***B.*** Felix, Hermann, Clemens ***C.*** Hermann, Clemens, Felix
D. Felix, Clemens, Hermann ***E.*** Hermann, Felix, Clemens

Part A: Each question is worth 3 points

1. Three children, Clemens, Max and Felix, enter a room one after the other. Max is not the first. Felix is not the first. Clemens is not the second. How many different orders are there for the children to enter the room?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 6

2. Four children, Paul, Mia, Felix and Leo, stand in a queue at the ice cream van, one behind the other. Paul is not the fourth. Leo is not the second. Felix is not the fourth. How many different orders are there for the children to stand in the queue?

A. 1 ***B.*** 6 ***C.*** 8 ***D.*** 10 ***E.*** 21

3. Three children, Sara, Tom and Lea, stand in a queue at the ice cream van, one behind the other. Tom is not the first. Lea is the first. In which order can they be?

A. Tom, Lea, Sara ***B.*** Tom, Sara, Lea ***C.*** Sara, Lea, Tom
D. Sara, Tom, Lea ***E.*** Lea, Tom, Sara

4. Four children, Emma, Ben, Lea and Clemens, stand in a queue at the ice cream van, one behind the other. Lea is somewhere before Emma. Emma is the fourth. Clemens is somewhere before Lea. In which order can they be?

A. Ben, Lea, Clemens, Emma ***B.*** Lea, Ben, Clemens, Emma ***C.*** Lea, Clemens, Ben, Emma
D. Ben, Clemens, Lea, Emma ***E.*** Clemens, Lea, Emma, Ben

Part B: Each question is worth 4 points

5. Four children, Lena, Hermann, Emma and Zoe, stand in a queue at the ice cream van, one behind the other. Emma is not the second. Emma is somewhere before Lena. Hermann is not the second. How many different orders are there for the children to stand in the queue?

A. 1 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

6. Five children, Leo, Paul, Sara, Anna and Lea, stand in a queue at the ice cream van, one behind the other. Paul is the second. Leo is the fifth. Anna is somewhere before Leo. How many different orders are there for the children to stand in the queue?

A. 6 ***B.*** 7 ***C.*** 8 ***D.*** 9 ***E.*** 12

7. Three children, Max, Clemens and Lena, stand in a queue at the ice cream van, one behind the other. Clemens is not the third. Clemens is somewhere before Lena. Max is somewhere before Clemens. Who is the first?

A. Nora ***B.*** Clemens ***C.*** Max ***D.*** Lena ***E.*** Zoe

8. Four children, Tom, Emma, Nora and Mia, stand in a queue at the ice cream van, one behind the other. Emma comes right after Mia. Tom is not the first. Emma is somewhere before Tom. Nora is somewhere before Emma. Who is the third?

A. Tom ***B.*** Ben ***C.*** Mia ***D.*** Emma ***E.*** Nora

Part C: Each question is worth 5 points

9. Three children, Nora, Felix and Lea, stand in a queue at the ice cream van, one behind the other. Felix is not the first. Lea is not the third. Which extra sentence makes sure that there is exactly **one** possible order?

A. Nora is somewhere before Felix.
B. Nora is not the first.
C. Felix comes right after Nora.
D. Felix is somewhere before Lea.
E. Felix is not the second.

10. Four children, Max, Lea, Emma and Ben, run a race. There are no ties. Emma is not the first. Max is not the first. Emma is somewhere before Lea. Which extra sentence makes sure that there is exactly **one** possible order?

A. Ben comes right after Max. ***B.*** Max is not the third. ***C.*** Max is somewhere before Emma.
D. Lea is not the third. ***E.*** Ben is not the third.

Answer Key

W1.	**B**	**W2.**	**C**	**W3.**	**D**	**1.**	**C**	**2.**	**D**	**3.**	**E**
4.	**D**	**5.**	**D**	**6.**	**A**	**7.**	**C**	**8.**	**D**	**9.**	**C**
10.	**C**										

Solutions

W1. B (2) — Write them all down: Sara, Felix; Felix, Sara. That is 2. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · w_all_orders]

W2. C (4) — There are 6 orders. In 2 of them Anna is the third. 6 – 2 = 4. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · w_one_rule]

W3. D (Felix, Clemens, Hermann) — Felix, Clemens, Hermann: Hermann is not first, Felix is not second, Clemens is not third. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · w_check_order]

1. C (2) — Write down all 6 orders, for example sorted by who is first. Then cross out every order that breaks a rule. 2 orders are left: Clemens, Max, Felix; Clemens, Felix, Max. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · not_rules]

2. D (10) — Write down all 24 orders, for example sorted by who is first. Then cross out every order that breaks a rule. 10 orders are left. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · not_rules]

3. E (Lea, Tom, Sara) — Check each order against every rule. Only Lea, Tom, Sara keeps all of them. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · which_order]

4. D (Ben, Clemens, Lea, Emma) — Check each order against every rule. Only Ben, Clemens, Lea, Emma keeps all of them. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · which_order]

5. D (5) — Go step by step: first use the strongest rule (a fixed place or 'right after'), then try the others in a system. Exactly 5 orders keep all the rules. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · four_children]

6. A (6) — Go step by step: first use the strongest rule (a fixed place or 'right after'), then try the others in a system. Exactly 6 orders keep all the rules. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · four_children]

7. C (Max) — Only one order keeps all the rules: Max, Clemens, Lena. So the first is Max. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · who_is]

8. D (Emma) — Only one order keeps all the rules: Nora, Mia, Emma, Tom. So the third is Emma. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · who_is]

9. C (Felix comes right after Nora.) — First list the 3 orders that keep the rules: Nora, Lea, Felix; Lea, Nora, Felix; Lea, Felix, Nora. Only the sentence "Felix comes right after Nora." leaves exactly one of them. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · one_more_rule]

10. C (Max is somewhere before Emma.) — First list the 3 orders that keep the rules: Ben, Max, Emma, Lea; Ben, Emma, Max, Lea; Ben, Emma, Lea, Max. Only the sentence "Max is somewhere before Emma." leaves exactly one of them. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · one_more_rule]